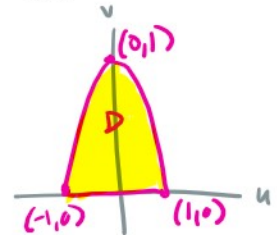


1. Let U and V be random variables with joint density $f(u, v) = \begin{cases} ku^2 & 0 \leq v \leq 1 - u^2 \\ 0 & \text{else} \end{cases}$

(a) Find k

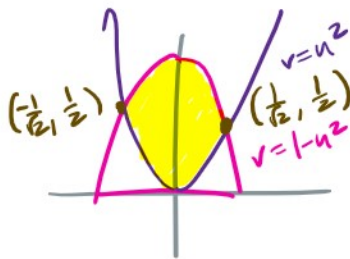
$$\begin{aligned} 1 &= \iint_D f_{uv} dA = \int_{-1}^1 \int_0^{1-u^2} k \cdot u^2 dv du = k \int_{-1}^1 u^2 v \Big|_0^{1-u^2} du \\ &= k \int_{-1}^1 u^2 - u^4 du = k \left[\frac{1}{3} u^3 - \frac{1}{5} u^5 \right] \Big|_{-1}^1 = k \cdot \left(\frac{2}{3} - \frac{2}{5} \right) \\ &= k \cdot \frac{4}{15} \rightarrow \boxed{k = \frac{15}{4}} \end{aligned}$$



(b) Find $E[V]$

$$\begin{aligned} E[V] &= \iint_D v f_{uv} dA = \int_{-1}^1 \int_0^{1-u^2} v \cdot k u^2 dv du = k \int_{-1}^1 u^2 \cdot \frac{1}{2} v^2 \Big|_0^{1-u^2} du \\ &= \frac{k}{2} \int_{-1}^1 u^2 (1 - u^2)^2 du = \frac{k}{2} \int_{-1}^1 u^2 - 2u^4 + u^6 du = \frac{k}{2} \left[\frac{1}{3} u^3 - \frac{2}{5} u^5 + \frac{1}{7} u^7 \right] \Big|_{-1}^1 \\ &= \frac{k}{2} \cdot \left(\frac{2}{3} - \frac{4}{5} + \frac{2}{7} \right) = \frac{15}{8} \cdot \left[\frac{70 - 84 + 30}{105} \right] = \frac{1}{8} \cdot \frac{16}{7} = \boxed{\frac{2}{7}} \end{aligned}$$

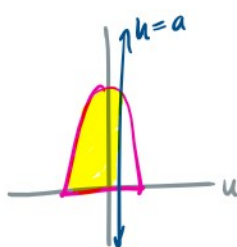
(c) Find $P(U^2 < V)$



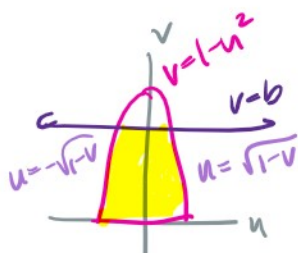
$$\begin{aligned} P(U^2 < V) &= \int_{-1/2}^{1/2} \int_{u^2}^{1-u^2} f_{uv} dv du = k \int_{-1/2}^{1/2} \int_{u^2}^{1-u^2} u^2 dv du \\ &= k \cdot \int_{-1/2}^{1/2} u^2 v \Big|_{u^2}^{1-u^2} du = k \int_{-1/2}^{1/2} u^2 (1 - 2u^2) du \\ &= k \int_{-1/2}^{1/2} u^2 - 2u^4 du = k \left[\frac{1}{3} u^3 - \frac{2}{5} u^5 \right] \Big|_{-1/2}^{1/2} = \frac{k}{15} (5u^3 - 6u^5) \Big|_{-1/2}^{1/2} \\ &= \frac{1}{4} \left[\left(\frac{5}{2\sqrt{2}} - \frac{6}{4\sqrt{2}} \right) - \left(-\frac{5}{2\sqrt{2}} + \frac{6}{4\sqrt{2}} \right) \right] = \frac{1}{2} \left[\frac{4}{4\sqrt{2}} \right] = \boxed{\frac{1}{2\sqrt{2}}} \end{aligned}$$

(d) Are U and V independent?

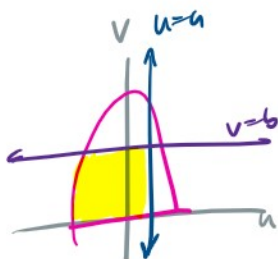
$$\underbrace{P(U \leq a)} \cdot \underbrace{P(V \leq b)} \stackrel{?}{=} \underbrace{P(U \leq a, V \leq b)}$$



$$\begin{aligned} P(U \leq a) &= \int_{-1}^a \int_0^{1-u^2} k u^2 \cdot dv du = k \int_{-1}^a u^2 v \Big|_0^{1-u^2} du = k \int_{-1}^a u^2 - u^4 du \\ &= k \left[\frac{1}{3} u^3 - \frac{1}{5} u^5 \right]_{-1}^a = \frac{15}{4} \cdot \frac{1}{15} (5u^3 - 3u^5) \Big|_{-1}^a = \frac{1}{4} [5(a^3+1) - 3(a^5+1)] \\ &= \frac{1}{4} [5a^3 + 2 - 3a^5] = F_U(a) \end{aligned}$$



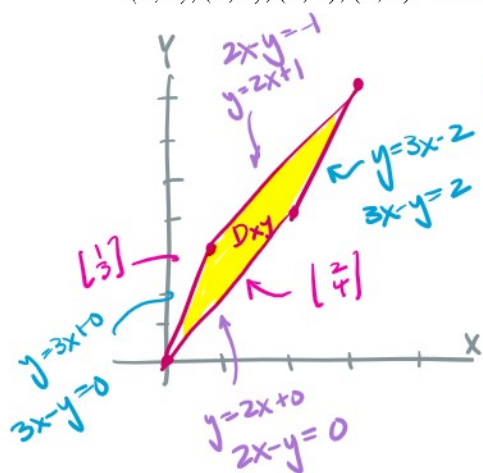
$$\begin{aligned} P(V \leq b) &= \int_0^b \int_{-\sqrt{1-v}}^{\sqrt{1-v}} k u^2 \cdot du dv = k \int_0^b \frac{1}{3} u^3 \Big|_{-\sqrt{1-v}}^{\sqrt{1-v}} dv \\ &= \frac{k}{3} \int_0^b 2(1-v)^{3/2} dv = \frac{2k}{3} \cdot \frac{(1-v)^{5/2}}{-5/2} \Big|_0^b = \frac{-4k}{15} [(1-v)^{5/2}]_0^b \\ &= -1 [(1-b)^{5/2} - 1^{5/2}] = 1 - (1-b)^{5/2} = F_V(b) \end{aligned}$$



$$\begin{aligned} P(U \leq a, V \leq b) &= \int_0^b \int_{-\sqrt{1-v}}^a k u^2 du dv = k \int_0^b \frac{1}{3} u^3 \Big|_{-\sqrt{1-v}}^a dv \\ &= \frac{k}{3} \int_0^b [a^3 + (1-v)^{3/2}] dv = \frac{k}{3} \left[a^3 v - \frac{(1-v)^{5/2}}{5/2} \right]_0^b \\ &= \frac{5}{4} \left[a^3 b - \frac{2}{5} (1-b)^{5/2} + \frac{2}{5} \right] = \frac{1}{4} [5a^3 b + 2 - 2(1-b)^{5/2}] \end{aligned}$$

NOT INDEPENDENT

2. Let X and Y be random variables with uniform density on the parallelogram with vertices $(0, 0)$, $(1, 3)$, $(2, 4)$, $(3, 7)$. Find $E[(2X - Y)(3X - Y)]$

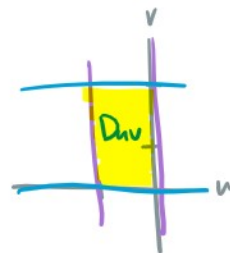


$$f_{XY}(x,y) = \begin{cases} \frac{1}{\text{AREA}(D_{xy})} = \frac{1}{2} & (x,y) \in D \\ 0 & \text{else} \end{cases}$$

$$\text{AREA}(D_{xy}) = \left| \det \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \right| = |-2| = 2$$

$$u = 2x - y \\ -1 \leq u \leq 0$$

$$v = 3x - y \\ 0 \leq v \leq 2$$



$$J_{uv \rightarrow xy} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = \begin{vmatrix} 2 & -1 \\ 3 & -1 \end{vmatrix} = 1$$

$$f_{uv}(u,v) = \frac{f_{xy}}{|J|} = \begin{cases} \frac{1}{2} \cdot \frac{1}{1} & (u,v) \in D_{uv} \\ 0 \cdot \frac{1}{1} & \text{else} \end{cases} = \begin{cases} \frac{1}{2} & (u,v) \in D_{uv} \\ 0 & \text{else} \end{cases}$$

$$E[(2X - Y)(3X - Y)] = E[uv] = \iint_{D_{uv}} uv \cdot f_{uv} \cdot dA = \int_{-1}^0 \int_0^2 uv \cdot \frac{1}{2} dv du$$

$$= \frac{1}{2} \int_{-1}^0 u du \cdot \int_0^2 v dv = \frac{1}{2} \cdot \left. \frac{u^2}{2} \right|_{-1}^0 \cdot \left. \frac{v^2}{2} \right|_0^2 = \frac{1}{2} \cdot (0 - \frac{1}{2}) \cdot (2 - 0) = \boxed{-\frac{1}{2}}$$

3. Let X and Y be discrete random variables with joint point mass function $p_{XY}(n, m) = C(n - m)$ for $1 \leq m \leq n \leq 100$.

(a) Find $P(N - M = 1)$

$$\begin{aligned}
 \text{Find } C: \quad 1 &= \sum_{n=1}^{100} \sum_{m=1}^n p_{XY}(n, m) = \sum_{n=1}^{100} \sum_{m=1}^n C \cdot (n - m) = C \cdot \sum_{n=1}^{100} \left[\sum_{m=1}^n n - m \right] \\
 &= C \cdot \sum_{n=1}^{100} \left[n^2 - \frac{n(n+1)}{2} \right] = C \cdot \sum_{n=1}^{100} \left[\frac{1}{2}n^2 - \frac{1}{2}n \right] = \frac{C}{2} \left[\sum_{n=1}^{100} n^2 - n \right] \\
 &= \frac{C}{2} \cdot \left[\frac{100 \cdot 101 \cdot 201}{6} - \frac{100 \cdot 101}{2} \right] = \frac{100 \cdot 101}{2} \cdot \frac{C}{2} \left[\underbrace{\frac{201}{3}}_{66} - 1 \right] = 25 \cdot 101 \cdot 66 \cdot C \\
 &\quad C = \frac{1}{25 \cdot 101 \cdot 66}
 \end{aligned}$$

$$\begin{aligned}
 P(N - M = 1) &= \sum_{n=1}^{100} p_{XY}(n, n-1) = \sum_{n=2}^{100} C \cdot (n - (n-1)) = \sum_{n=2}^{100} C = 99C \\
 &= \frac{99}{25 \cdot 101 \cdot 66} = \boxed{\frac{3}{50 \cdot 101}}
 \end{aligned}$$

(b) Evaluate $E[N | N - M = 1] = \sum_{n=1}^{100} n \cdot P(N=n | N-M=1)$

$$\begin{aligned}
 &= \sum_{n=2}^{100} n \cdot \frac{C(n - (n-1))}{\frac{P(N=n, M=n-1)}{P(N-M=1)}} \\
 &\quad \text{300}
 \end{aligned}$$

$$= \frac{101 \cdot 50 C}{3} \cdot \sum_{n=2}^{100} n = \frac{101 \cdot 50 \cdot C}{3} \cdot \left(\frac{100 \cdot 101}{2} - 1 \right)$$

$$\boxed{= \frac{101 \cdot 50}{3} \cdot \frac{1}{25 \cdot 101 \cdot 66} \cdot (5049)}$$

$$= \frac{1}{99} \cdot 5049 = \frac{561}{11} = \boxed{51}$$

4. What is the mean of the continuous random variable satisfying $P(X \leq \alpha) = 1 - e^{-\alpha^2}$ for all $\alpha \geq 0$?

$$\text{density: } f_X(\alpha) = \frac{d}{d\alpha}(\overbrace{F_X(\alpha)}) = \frac{d}{d\alpha}(1 - e^{-\alpha^2}) = 2\alpha e^{-\alpha^2} \quad (\alpha \geq 0)$$

$$\begin{aligned} E[X] &= \int_0^{\infty} t \cdot f_X(t) dt = \int_0^{\infty} t \cdot 2t e^{-t^2} dt \\ &= 2 \int_0^{\infty} t^2 e^{-t^2} dt = \int_{-\infty}^{\infty} t^2 e^{-t^2} dt \\ &= \sqrt{\pi} \cdot \frac{1}{2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{\pi} \cdot \frac{1}{2}} t^2 e^{-\frac{t^2}{2 \cdot \frac{1}{2}}} dt = \sqrt{\pi} \cdot \text{Var}(N(0, \frac{1}{2})) \\ &= \sqrt{\pi} \cdot \frac{1}{2} = \boxed{\frac{\sqrt{\pi}}{2}} \end{aligned}$$